মেট্রোপলিটন ইউনিভার্সিটি
Metropolitan University

A Thesis Report

Submitted To

The Department of Computer Science and Engineering, Metropolitan University, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

On

Hybrid Fusion Framework for Human- and AI-Generated Misinformation Detection

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August 2026

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STUDENT'S DECLARATION

We hereby declare that the work presented in this thesis, titled "**Hybrid Fusion Framework for Human- and AI-Generated Misinformation Detection,**" is the outcome of our own research carried out under the supervision of **Samia Rahman Rima**. To the best of our knowledge, this thesis contains no material previously published or written by another person, except where due reference is made in the text. This work has not been submitted, in whole or in part, for any other degree or diploma at this or any other institution.

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Certification

Thus, we hereby state that the thesis entitled " **Hybrid Fusion Framework for Human- and AI-Generated Misinformation Detection**" has been done under our guidance by

1. MD. Jeshan Khan
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This thesis is completely original and, according to our best knowledge, fulfills the requirements of the

concerned degree program

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Abstract

The rapid spread of misinformation on social media has become a very critical and challenging situation nowadays. False information frequently disseminates widely before detection and intervention can occur. Existing misinformation detection methods primarily focus on identifying false content after it has already gained significant traction, while most early detection methods treat misleading or misinformation as a single category without comparing between human-authored and AI-generated content. This limitation has become very important with the emergence of large language models. This capability allows for the prediction of highly realistic AI-generated misinformation at an unprecedented scale. To address these challenges, this paper proposes a hybrid multimodal framework that integrates both content and image information for early misinformation detection and attribution. In parallel, a temporal graph neural network models the evolving propagation graph to capture temporal diffusion dynamics during the early stages of information spread. Finally, a late fusion mechanism combines the output of both branches to perform three- or four-way classification in a single model. The proposed framework is expected to provide faster and more fine-grained information detected which is false or misleading to people while achieving measurable improvement in early-stage F1-score compared with existing content-based methods.

Keywords: misinformation detection, fake news, multimodal fusion, graph neural networks, temporal propagation modelling, DistilBERT, GoEmotions, stacked generalisation, early rumour detection.

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Acknowledgement

First of all, we would like to thank our advisor, **Samia Rahman Rima**, for her valuable advice throughout our research process. Her valuable criticism, especially in relation to our research problem definition and model evaluation, was a crucial component of our success. Secondly, we would like to thank the professors of the Department of Computer Science and Engineering for their lessons, which became the basis for this research. Lastly, we would like to thank the open source community and authors of the pre-trained models and datasets used in our study, such as **DistilBERT** and **GoEmotions** from Hugging Face Transformers, and the dataset of the AI-generated image and news articles used in this thesis.

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Chapter 1: Introduction

Background and Motivation

Social media, instant messaging, and news websites have revolutionized the way information is generated and disseminated. Millions can receive the information in a matter of minutes, but the information disseminated through these platforms can contain a lot of misinformation. Misinformation and disinformation have the potential to negatively impact elections, public health, fraud, and institutions' credibility. Until now, research on misinformation has focused on three kinds of human-created content that contain falsehood: false reports, satirical content taken as factual, and selectively framed reports.

However, in the time of Generative AI, large language models generate human-like text, and diffusion image generators generate realistic-looking fake images. This is why the whole area of research has been reoriented to explore this new pathway of information generation. These kinds of detection techniques, which use typical characteristics of human writing, such as awkward expressions and grammatical mistakes, are not useful in detecting this new kind of AI-generated content.

The other problem with contemporary research is that most models operate on a single signal type without accounting for multiple types simultaneously. The text classifiers don't pay attention to the images, the image forensic tools ignore the text part, and neither of them takes the spreading process into account. However, the research shows that each of these factors- the text features, whether the images are genuine, and the spreading and speed of spreading- is capable of predicting the veracity or fake nature of the news separately. The combination of these signals, rather than reliance on one of them, would improve the performance of the tool. For instance, emotional exploitation can be considered as a factor that proves the detection of emotions and can help detect misinformation. The credibility of the publication source is an indicator of the credibility of the news itself. Network spreading process characteristics are the attributes of the content that cannot be observed. In terms of AI detection tools, tools such as DetectGPT, FastDetectGPT, and GPTZero show that it is possible to estimate the probability of the text being artificial on the basis of its statistical features.

Problem Statement

This research work tackles the problem of combining various types of signal related to the news article in question, both while the article is being created and even after its publication and propagation, into a single model that (a) can detect the fakeness or reality of the article, (b) works efficiently regardless of the nature of the article content whether it has been generated by a human or a machine learning algorithm, and (c) uses the propagation context signal of the article if such

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information is available. The problem here is inherently multimodal and temporal classification: it involves combining the static signal of the article (the text and, possibly, image) along with the temporally dynamic graph propagation information in order to classify the article into one of two classes.

Research Objectives

This thesis sets out to:

● Create a pipeline for extracting textual, emotional, source-credibility, and AI-generation features from news article text.

● Create a comparable pipeline for estimating the AI-image-generation likelihood of images alongside descriptions.

● Model how information spreads with a Temporal Graph Transformer that relies on attention over ordered cascade nodes according to their occurrence.

● Design and develop a Hybrid (Early + Late) Fusion architecture that integrates text, image, and propagation embeddings for the construction of a single end-to-end trainable classifier.

● Evaluate the proposed framework against content-only baselines and sequence/graph baselines (LSTM, CNN, static GNN), and test its ability to detect misinformation early, using only a portion of the observed propagation cascade.

● Evaluate the pros, cons, and current limitations of the framework, and propose specific suggestions for future development.

Scope of the Study

The current thesis tackles the binary classification task (fake vs. real) for news articles in the English language. This task involves analysis using the propagation cascade created automatically between articles, based on publicly available metadata per article, rather than the more reliable method of creating a single live social media stream, which is described more extensively in Chapter 6. In this case, the image detector generated by AI and the DetectGPT-lite text detector can be used as feature extractors; however, they should not be used as classifiers because of the ground truth (fake/real) labels in the dataset.

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Significance of the Study

This research work contributes to the ever-growing body of literature surrounding multimodal misinformation detection in several distinct manners. It presents a framework—being among the relatively few within current literature—that fuses text, image, emotion, source credibility, AI-generation likelihood, and propagation dynamics into a single trainable architecture, instead of treating these as separate signals that are combined only after the fact. The Time-Aware Graph Transformer is introduced specifically for misinformation cascades, where each node's time of appearance is used directly as part of the attention mechanism. This allows for the framework to be evaluated for early detection; a capability with real practical value, since the harm caused by misinformation tends to grow with how far it spreads before anyone intervenes. The study provides a transparent and reproducible account of results, which were obtained on a modestly sized dataset. An honest discussion regarding where the current implementation falls short is included, which should be of direct use for future researchers who are looking to scale this type of model. Furthermore, it offers insights that are valuable for those who want to expand upon this work in the future. More broadly, the main contributions of this work can be summarised as: A hybrid feature-fusion framework that combines semantic, emotional, AI-detection, source-credibility, and temporal propagation signals.

• A single deep learning model capable of detecting both human- and AI-generated misinformation.

• The integration of DistilBERT, GoEmotions, DetectGPT/FastDetectGPT, source credibility features, and temporal propagation modelling within one detection pipeline.

• An evaluation of the framework against standard benchmark datasets and metrics (accuracy, precision, recall, F1-score).

• A framework intended to be efficient and scalable enough to support future misinformation detection systems on real-world digital platforms.

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Thesis Organisation

The remainder of this thesis is organised as follows.

Chapter	Content
Chapter 2	Reviews the literature on human- and AI-generated misinformation detection, multimodal fusion methods, and graph/propagation-based detection approaches.
Chapter 3	Describes the datasets, class-balancing procedure, and feature-extraction methodology for text, emotion, source credibility, and AI-generated images.
Chapter 4	Presents the system design: propagation-graph construction, the Time-Aware Graph Transformer architecture, the regularised cross-validated propagation model, and the leak-free stacked hybrid classifier.
Chapter 5	Reports the experimental results: content-only baselines, leak-free hold-out evaluations, propagation-only results, baseline comparisons, the stacked hybrid model, an early-detection evaluation, and a feature-correlation analysis.
Chapter 6	Discusses the findings, interprets the observed model behaviour, and honestly addresses the remaining limitations of the corrected pipeline.
Chapter 7	Concludes the thesis and outlines directions for future work

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Chapter 2: Literature Review

i. Text-Based Fake News Detection

Early attempts for automated falsified data detection were based on manually designed features like n-grams, parts-of-speech tagging, and punctuation. News classification traditionally used classical machine learning approaches like Support Vector Machines (SVM) and Naïve Bayes. The key improvement came when researchers started combining distributional representations and sequence models such as LSTM, GRU, and CNNs. The most important breakthrough in the field of machine learning happened when the transformer architecture, which includes a self-attention module and pretraining techniques used in BERT, was introduced. Among these models, BERT is considered one of the top models that performs well in many NLP tasks. However, training BERT requires high computational resources. That is why the variant of BERT called DistilBERT, which retains 97% of the performance of BERT with only 60% of its parameters, was chosen as the base model. There are studies which compare transformer models and autoregressive large language models (Raza et al., 2024); benchmarking across different models and datasets has been done (Pérez-Rosas et al., see arXiv:1905.04749). These results show that encoder-only transformer models can be used as a baseline. Concerning the dataset design described in chapter 3, there are two benchmark datasets used: LIAR (Wang, 2017) and FakeNewsNet (Shu et al., 2018).

ii. Emotion and Affective Signals in Misinformation

There has been an abundant amount of studies on the issue that misinformation is not neutral in terms of emotions. In one study where researchers analyzed large sets of tweets from Twitter, Vosoughi, Roy, and Aral (2018) found out that false stories were diffused much more efficiently than true stories. Moreover, the researchers found out that false stories caused more surprise and disgust in the replies and true stories caused more trust and anticipation. This means that emotional reactions are part of diffusion process dynamics and not just its consequence. Chuai and Zhao (2022) found out that the anger expressed by a post of fake news is positively correlated with diffusion ease of this post regardless of the platform where it is shared. This finding was confirmed by results of a cross-platform study conducted by McLoughlin et al. which showed that misinformation sources cause much more outrage than trustworthy sources. Moreover, outrage and not accuracy is what predicts resharing. Some additional studies about emotional sequences of disinformation showed that false content follows some particular emotional sequence (e.g., surprise → fear/anger → a call to action) and it is done in order to maximize users' engagement before critical thinking. Given all the information above, in this thesis the author uses GoEmotions (Demszky et al., 2020), which is the largest fine-grained manually

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annotated emotion dataset (58k Reddit comments and 27 emotion categories including neutral) in order to build an emotion classification part of the feature pipeline.

iii. Detecting AI-Generated Text

While the ability of LLMs to produce fluent human-like text has been widely recognized, there has been a line of research aimed at detecting the machine origin of the text in terms of being able to distinguish machine-generated text from human-written text without using any information related to the training dataset of the source LLM. The DetectGPT algorithm proposed by Mitchell et al. (2023) represents one of those approaches which operates in the zero-shot setting and is based on the assumption that the text sampled from LLM belongs to the region of negative curvature of the log-probability function of the source model. As a consequence, it suffices to apply some perturbations to the sample and calculate log-likelihoods in order to detect its machine origin. In order to tackle the high computational cost of the perturbation step in the DetectGPT algorithm, Bao et al. (2023) proposed an alternative sampling-based approach to estimating conditional probability curvature and demonstrated similarly good or even better results with a lower resource consumption. The DetectGPT-lite detector implemented in this thesis is a direct result of this research. In turn, Chen and Shu (2023) provide empirical evidence supporting the claim that the misinformation produced by LLMs poses a different and potentially more difficult threat in comparison to the human-created one. For a number of classes of misinformation, it was shown that both human annotators and automated detectors had a harder time distinguishing LLM-generated misinformation from the human-written one with the same semantics. This result is placed in the context of fighting misinformation in the age of LLMs in Chen and Shu (2024) review and serves as motivation for the approach to misinformation filtering proposed in this thesis.

iv. Detecting AI-Generated Images

On the other hand, there is another line of forensic research dedicated to the problem of identifying if an image has been created with the use of some generative model rather than taken through the camera. In one of the most cited papers, Wang et al. (2020) proved that the detector trained on images produced by a single generator CNN-based (ProGAN) achieves surprisingly high levels of generalization for other GAN architectures; hence, there were some systematic artefacts in images created with the use of CNNs at that time. Barni et al. (2020) further improved the effectiveness of their detection technique for faces produced with the help of the GAN network by exploiting the inconsistency of the cross-band co-occurrence statistics of color channels that were harder for the generator to imitate than the spatial texture. The emergence of diffusion models was another challenge for forensics studies due to the fact that diffusion images did not have any characteristic

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artefacts that GAN images had: Corvi et al. (2022) explored the problem of detecting diffusion model images and proved that the detectors aimed to exploit GAN artefacts generalize badly; hence, there was a need for creating architecture-agnostic forensic features. Another recent study that exploits the idea of global-local feature fusion in order to solve the problem of detecting AI-synthesized images (arXiv:2211.08615) continues the mentioned line of works and seeks to create detectors generalizable to both generators.

v. Source Credibility Modelling

No matter what the article is about, its source is a predictor already. According to the credibility-based approach by Zhou et al., not only is the source history of producing falsehoods a significant factor that affects the validity of an article but also structural features such as the amount of listed authors. The abovementioned point of view aligns with the FakeNewsNet dataset (Shu et al., 2018), which takes source-level metadata, social context, and spatiotemporal diffusion together with content into account since content alone is insufficient to detect fake news – a good fake article can be written quite well, whereas a genuine article can be of poor quality. In general, a literature overview on the subject (Zhou & Zafarani, 2020) mentions evaluating the credibility of a source as one of the three major paradigms of detecting fake news (the other two are content-based and social context-based). Additionally, sources emphasize the importance of the source signal in the early detection of fake news because it is available prior to the spreading of the article. That is why the source credibility should be included in the fused feature vector discussed in Chapter 3.

vi. Propagation-Based and Graph Neural Network Approaches

Beyond static content and source features, a separate line of research models how a story spreads through a network over time, on the premise that false and true information propagate differently. Bian et al. (2020) introduced Bi-GCN, a bi-directional graph convolutional network that operates on both the top-down (information cascade) and bottom-up (source-tracing) versions of the propagation tree, and showed that combining the two directions substantially improves early rumour detection over content-only baselines. Related work on dynamic graph convolutional networks with an attention mechanism represents the propagation structure as a sequence of evolving snapshots so that the model can weigh more recent, structurally central interactions more heavily when scoring a story, while a recurrent graph convolutional formulation folds the temporal ordering of replies directly into the message-passing step rather than treating each snapshot independently. Rossi et al. (2020) generalised this idea beyond misinformation detection with Temporal Graph Networks (TGNs), a memory-augmented encoder for continuous-time dynamic graphs that updates a per-node memory vector as new time-stamped interactions arrive; the temporal encoding scheme used for the propagation branch of this thesis (Section 4.2) is directly inspired by this line of work, in particular its use of a

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learnable, bounded time-embedding rather than a raw timestamp. Taken together, these studies establish that (i) directionality (who reacted to whom) and (ii) time (how quickly and in what order reactions occurred) both carry independent diagnostic signal beyond the text of the story itself, and that this signal is most informative in the early stages of a cascade, before the story has had time to fully propagate — which is precisely the regime this thesis targets with its early-detection evaluation (Section 5.8). A limitation shared by all of the above is that they were developed and validated on real, observed cascades (e.g. Twitter15/16, PHEME, Weibo); Chapter 6 revisits this point when discussing the synthetic-cascade limitation of the propagation branch proposed here.

vii. Multimodal Fusion Strategies

Since no single signal category is sufficient on its own, a substantial body of work fuses two or more of them into a single classifier. EANN (Wang et al., 2018) was among the first to combine textual and visual features for fake-news detection, adding an event-discriminator branch trained adversarially so that the shared representation generalises to unseen events rather than memorising event-specific cues. GAME-ON (2022) fuses textual and visual modalities through a graph-attention mechanism operating at the level of within-article entities and image regions rather than a single pooled vector per modality, allowing the model to attend to the specific claim-image correspondences that make a story credible or not. GS2F (Guided Semantic Fusion) extends this further by using graph structure to guide how semantic (text) and visual features are combined, reporting consistent gains over unimodal and naively-concatenated multimodal baselines. On the forensic side, detecting whether an accompanying image is itself AI-generated has become its own sub-field: Wang et al. (2020) showed that CNN-based GAN detectors trained on a single generator generalise surprisingly well to unseen ones, Barni et al. (2020) improved face-forgery detection using cross-band colour-channel co-occurrence statistics, Corvi et al. (2022) showed that these GAN-era detectors transfer poorly to diffusion-model images and called for architecture-agnostic forensic features, and the Global-and-Local Feature Fusion (GLFF) approach (2022) combines global image statistics with local patch-level cues to generalise across both GAN and diffusion generators. Across this literature, the consistent finding is that fusing even two signal categories — most often text and image — reliably outperforms either signal alone, which directly motivates extending fusion to a wider set of signal categories, as attempted in this thesis (Section 2.9).

viii. Human vs. AI-Generated Misinformation: An Emerging Research Gap

According to recent disinformation detection research, past studies have generally overlooked the possibility that the disinformation of high quality could be generated by artificial intelligence because of their assumptions about high-quality disinformation being created by humans. However, there have been some studies to challenge the underlying

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assumptions. One example is the work of Chen and Shu (2023) where the authors created a taxonomy of disinformation created by language models and compared the effectiveness of automatic detection against human detection of new human-generated disinformation. In addition, Chen and Shu (2024) analyzed the disinformation detection problem in relation to countermeasure technologies.

ix. Summary and Positioning of This Thesis

As a whole, the reviewed literature presents a classification of five major categories of misinformation signals into: text/semantics (Section 2.1), affective (Section 2.2), AI generation probability of the text and images (Sections 2.3-2.4), source credibility (Section 2.5), and structural/propagation properties, including their temporal dynamics (Section 2.6). So far each category has been independently proposed and evaluated. Multimodal fusion approaches (Section 2.7) show that even fusion of only two out of the five categories—text and image—constantly outperforms respective unimodal baselines. However, only very few works propose to fuse more than two of the above categories within an evaluated end-to-end pipeline, and even fewer attempt at addressing the problem of distinguishing human and AI authorship stated in Section 2.8. It is in this gap that the Hybrid Fusion Framework proposed in this thesis seeks to fill by integrating all five signal categories into a single trainable framework and evaluating it according to the protocol of baseline comparison, propagation model cross-validation, leak-proof stacking, and early-detection analysis over different observation windows. The protocol follows well-established experimental protocols in propagation-based detection literature (Bian et al., 2020; Rossi et al., 2020).

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Chapter 3: Datasets and Feature Engineering Methodology

1. Overview of the Data Pipeline

Three types of raw data go into the pipeline. First up is a set of news articles divided into four CSV files, two with fake stories and two with real ones, labeled as file pair 1 and file pair 2 for each type, that together make up the supervised classification dataset. The second group is AI-made pictures covering eight object types and is used only to set up the standalone AI image detector from Section 3.7, and the third group is three large, general GoEmotions-style Reddit comment files of 70,000, 70,000, and 71,225 rows that were saved for hand-checking only and not automatically added to the main article features because their row totals don't line up with any article category. We download the raw files just once from an organized Google Drive folder and keep them saved locally, so later steps never have to download them again.

So the big change here is how we're handling all the features for each article. Now, everything - the text embedding, emotion score, credibility score - gets calculated and added as a new column to one big dataframe per article. We use an article_id to keep track of everything. Before, we were doing things differently. The text embedding part would create three separate rows for each article, while the credibility part would just have one row. Then, when we combined all these features, we were doing it based on the row position, not the actual article identifier. Which, looking back, was a recipe for disaster - it would sometimes match one article's embedding with a completely different article's label. No wonder the accuracy was so bad, like 10-41%. But now, since we're building one dataframe per article from the start, this kind of mix-up can't happen. Every feature gets added to the same dataframe, so it's all tied together properly.

Dataset Composition

Data Type	**Count**
Images (AI-generated, 8 object types)	**24877**
Text data (news articles + Reddit comments)	**234421**
Total dataset	**259298**

2. Class Balancing Before Feature Computation

So we've got this big pool of articles - 5,755 labelled as fake and 17,441 labelled as real. Before we start doing anything with them, like figuring out what they're about or how people might feel about them, we need to balance them out. We want 5,000 of each, and we try to spread them out evenly across the different files they came from. It makes sense to do this now, rather than later, because otherwise we'd be wasting time on articles that are just going to get thrown out anyway. And by doing it this way, we know exactly what we're working with - no guessing. There are tables that show the original numbers and what we ended up with after sampling.

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Class	File pair #1	File pair #2	Total raw rows
Fake News	5,323	432	5,755
Real News	16,817	624	17,441

Table 3.1: Raw article counts by class and source-file pair, before equalization

Class	**Sampled from pair #1**	**Used from pair #2 (all available)**	**Total sampled**
Fake News	4,568 / 5,323	432 / 432	5,000 / 5,000 (target met)
Real News	4,376 / 16,817	624 / 624	5,000 / 5,000 (target met)

Table 3.2: Class-equalisation outcome (target 5,000 Fake + 5,000 Real articles). File pair #2 for both classes had fewer than the 2,500-per-pair target available, so all of its rows were used and the shortfall was made up from file pair #1.

3. Text Preprocessing

So we took the article titles and cleaned them up. We made everything lower case, got rid of punctuation and any weird characters, removed common words like "the" and "and", and simplified the words to their base form - like "running" and "runs" both just become "run". We stored the cleaned titles in a new column, so we still have the originals if we need to check something. But if a title ended up being empty after all that, we just got rid of it. That meant we lost a few articles - 5 out of 10,000, which isn't a lot, only 0.05%. So we went from 5,000 fake and 5,000 real articles to 4,999 fake and 4,996 real ones.

4. Text Feature Extraction — DistilBERT Embeddings

So we took the title of each article and cleaned it up. Then we ran it through this AI model called distilbert - it's a pre-trained one. We used a library to access it, makes things easier. What we got out of it was like a fingerprint for each article - a bunch of numbers that represent the article. It's 768 numbers, to be exact. We did this in batches, using a GPU to speed things up, and saved all these fingerprints to a file. We made sure they were lined up right, so the first article in a group matches up with its fingerprint. This was important because we had some issues before where things got mixed up. Anyway, we ended up with almost 5,000 of these fingerprints for fake news and real news each.

5. Emotion Feature Extraction — GoEmotions

So we used this pretrained model to score each article for 28 different emotions - including neutral, just to cover all bases. We took those scores and added them as new columns to our main dataset, so now each article has a bunch of extra info. We did this for both the fake news and real news articles - about 5,000 of each. I guess it's worth noting, we used to have a lot more emotion data - 59 columns worth. But it turned out all those extra stats didn't really add much value, so we simplified things and just stuck with the 28 basic emotion scores. Oh, and we also had these huge

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files of Reddit comments - tens of thousands of them. Each comment had a ton of info, like the text, which subreddit it was from, and how people rated the emotions in it. But the thing is, the number of comments didn't match up with the number of articles we were working with, so we just kept them on the side for manual checking. We didn't actually use them to train our model or anything.

6. AI-Generated-Text Detection Score (DetectGPT-lite) — Disabled by Default

So we have this thing called DetectGPT-lite - it's like a simpler version of DetectGPT. We can turn it on or off with a flag, and for the stuff in this thesis, we actually turned it off. The reason is, when we first tried it out, it just gave us a score of 0 for every article, which wasn't helpful at all. Plus, it was slowing things down. So we left it out of the main results, but it's still there if someone wants to use it in the future - once we figure out how to make it work properly, that is.

7. Source Credibility Feature

Each article gets a rough trust score from 0 to 1 after we pull its domain from the URL, tidy it up, and check it against a short list of known reliable and unreliable websites. The pipeline's old domain regex only caught URLs starting with https://, so it missed about 99% of real-world links, but now it works whether the scheme is present or not. After the fix, the share of rows with matching domains climbed from about 1% to 4.8% for fake news and 7.9% for real news. Any article that can't be matched to a trusted domain automatically gets a neutral 0.5, and while Chapter 6 flags the still-low match rate as a limitation, it covers five to eight times more articles than the earlier broken version did.

8. AI-Generated-Image Detection Score (Independent Task)

We ran a pretrained Hugging Face vision transformer named umm-maybe/AI-image-detector, which classifies images as either AI or human-made, on the 1,050 images across eight categories in the AI_Generated_Image collection to see how likely each one was AI-generated. Since the news datasets don't actually link any specific images to specific articles, we score and report images completely separately from the article truth-checking system and don't add those scores into the article data table, which fixes an old mistake where we had randomly assigned an image score to each article even though no real connection existed. For each category, Table 3.3 shows the average AI score and the share of images marked as fake with a score above 0.5.

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Category	Images Scored	Mean AI-Likelihood Score	% Flagged as AI-Generated
Airplane	50	0.936	100.0%
Animal	750	0.405	39.6%
Car	50	0.826	100.0%
Flower	50	0.429	38.0%
Fruit	50	0.190	0.0%
Human	50	0.258	12.0%
MotorBike	50	0.537	50.0%
Person	50	0.667	78.0%

Table 3.3: AI-generated-image detection score summary by object category (independent task).

So the flagging rates still vary a lot depending on the category - like, nothing gets flagged for Fruit, but pretty much everything gets flagged for Airplane and Car. That suggests the detector isn't really consistent across different types of objects. This is actually something we talk about more in Chapter 6 as a limitation. But the good thing is, since we're not tying the image task to article classification anymore, this issue doesn't affect the results we got for detecting false info, which we cover in Chapter 5.

9. Feature Fusion — Article-Level Representation

The Fake_News and Real_News datasets were merged into a single corpus of 9,995 articles with a near-balanced class distribution, after which each article's text was converted into a dense 768-dimensional embedding (e.g., via a pretrained transformer encoder) to capture its semantic content. To reduce dimensionality while retaining signal, Principal Component Analysis (PCA) was applied to these 768-dimensional vectors, projecting them onto 100 orthogonal components that preserved approximately 83% of the total variance in the original embedding space — this step reduces computational cost and mitigates the curse of dimensionality for downstream classifiers. These 100 PCA-derived components were then combined, via a feature fusion (concatenation) step, with a set of hand-crafted auxiliary features — including emotion probability scores (from a sentiment/emotion classifier) and a credibility score (a separately computed metric reflecting source or content trustworthiness) — while two additional candidate scores were excluded due to reliability or availability issues, yielding a final fused feature matrix of 129 dimensions per article; this produces a sample-to-feature ratio of roughly 77.5:1 (9,995 samples / 129 features), which is favorable for reducing overfitting risk in the subsequent classification models.

Feature Block	Dimensionality	Source
DistilBERT title embedding (PCA-reduced)	100	Section 3.4, reduced via PCA (82.86% variance retained)
GoEmotions class probabilities	28	Section 3.5
Source-credibility score	1	Section 3.7
Total fused feature vector	129	—

Table 3.4: Composition of the final 129-dimensional fused article-level feature vector.

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10. Train-`Test Splitting Strategy

The framework is tested at multiple levels of granularity, all using a fixed random seed (42) and stratification where relevant (see Chapter 5 for full details):

- Article-level, 5-fold cross-validation: Full 9,995-article dataset; used for content-only baseline classifiers (Section 5.1).
- Article-level, 70/30 hold-out: Leak-free split (6,996 train / 2,999 test); PCA and scaling fitted on training data only (Section 5.2).
- File-pair level, 70/30 hold-out: Leak-proof splits applied separately per source — file pair #1 (6,125/2,625) and file pair #2 (603/259) (Section 5.3).
- Cascade-level, 70/15/15 split: 1,200 propagation cascades divided into 840 train / 180 validation / 180 test, used for initial TGNN training and convergence diagnostics (Section 5.4).
- Cascade-level, 5-fold cross-validation: Same 1,200 cascades, evaluated with a more regularized model to obtain out-of-fold predictions for stacking (Section 5.5).
- Cascade-level, primary 70/15/15 train/validation/test split, repeated 5 times: Same 1,200 cascades; content, propagation, and hybrid are all scored on identical splits within each repeat, with model/hyperparameter selection restricted to the validation portion and a single scoring pass on the held-out test portion (Section 5.12).
- Cascade-level, 10-fold cross-validation: Same 1,200 cascades, re-run independently at 10 folds as a robustness check on the 5-fold results above (Section 5.13).

All splits use a fixed random seed (42) for reproducibility, and stratification is applied wherever class balance across a split is material to the evaluation.

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Chapter 4: System Design and Implementation

1. Multimodal Temporal Propagation-Graph Construction

As there is no actual propagation trace for any of the articles, a synthetic one is created for every article, where each node of the graph represents a single share/react event. Synthetic cascades are produced according to the finding by Vosoughi, Roy and Aral (2018), stating that false news tends to be disseminated faster and further than true news. For Fake-labelled articles, the branching factor distribution is set to be higher (2.5–4.0) and the average inter-event delay lower (2–15 minutes) than for Real articles (branching 1.2–2.0; delay 20–90 minutes). The content vector of each node is constructed as a 129-dimensional vector representing fused features of the corresponding article (Section 3.9) plus Gaussian noise, since no pairing between images and articles and therefore no image score can be included in the content vectors of the cascades – a correction from the initial version of the pipeline, which was erroneously doing so. Additionally, three structural features are concatenated to the content vector of each node, resulting in node feature vectors of 132 dimensions.

The essential fix to the previous version of the pipeline involves dealing with the problem of the node timestamp. The raw timestamp values in minutes (up to 10,000) were being passed to an unbounded linear term in the Time2Vec temporal encoding, resulting in exploding training losses (into the tens of thousands) and an effectively collapsed propagation model predicting one class for all inputs. In the new pipeline, the time since the root of the cascade is z-scored with respect to each cascade independently (hours instead of minutes with respect to the mean and standard deviation of the particular cascade), and then this normalised elapsed time is passed to the Time2Vec encoding, while the LayerNorm is added to the raw node features before the first projection layer. These modifications ensure that all inputs to the temporal-attention mechanism are on a bounded and properly scaled range.

Since the calculation of an entire 30 node cascade for all 9,995 equalized documents is computationally heavy, the maximum number of cascades to be created for each category is limited to 600 (MAX_ARTICLES_PER_CATEGORY_GRAPH = 600), that is 600 Fake and 600 Real document cascades or 1,200 propagation graphs in total which is four times the 300 graph dataset in the previous version of this pipeline. This graph data set is used for single split analysis as well as the 5 fold cross-validation propagation model.

2. Time-Aware Graph Transformer (TGNN) Encoder

First, the structure/time encoder is a specially designed Time-Aware Graph Transformer (TGNNClassifier). Input 132-dimensional node features are first normalised with a LayerNorm layer (the fix from Section 4.1) and linearly projected to the 64-dimensional model width (d_model = 64). Single temporal graph-attention layer (n_layers = 1, n_heads = 2) calculates multihead attention for nodes of each cascade, in which the raw attention scores are enhanced with (i) time

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bias computed from the pairwise (per cascade) normalised time difference between nodes through Time2Vec (time_dim = 16) and a linear mapping to per-head biases, and (ii) structural bias based on whether a pair of nodes is connected in the cascade graph. Attention is masked to ensure that a node attends to other nodes only of the same cascade. Residual connections and LayerNorm are applied after each attention block, with subsequent application of two-layer feed-forward network, following Transformer block standard structure. Finally, per cascade mean-pooling of node embeddings is used as an input to the small classification head (Linear → ReLU → Dropout → Linear) that returns the Fake/Real label prediction. The number of trainable parameters in this model is 44,464.As an initial test of training stability, one-time, one-split experiment with training TGNNClassifier was conducted on the 840-cascade training split (see Section 3.10) using class-weighted cross-entropy loss and Adam optimiser, with early stopping on validation F1 score (with patience 8 epochs and maximum number of 40 epochs). Accuracy on training data steadily increased during epochs (from approximately 79.5% in epoch 15 to 86.9% in epoch 25), whereas validation accuracy stabilised in the range of 72–74% with the best F1 score of 0.7390; the training process was terminated early in epoch 29 when F1 score of validation stopped to grow, hence the training had converged and did not face problems of loss divergence and majority class collapsing observed with the previous, unnormalised version of the pipeline. Test-set results of this diagnostic experiment are shown in Section 5.4.

3. Regularised Propagation Model for Cross-Validation (TGNNClassifierV2)

A split of only 1,200 cascades, i.e. a 840/180/180 split, serves a useful stability check on its own, but it can make the model appear to be performing better or worse than expected just by luck. In order to get a robust, variance-aware estimate of the model's propagation-only performance, and to provide leak-free out-of-fold predictions for hybrid stacking (Section 4.4), a more modest sibling architecture of the main model, TGNNClassifierV2, is considered using stratified 5-fold cross-validation of all 1,200 cascades. TGNNClassifierV2 restricts the width of the hidden layers to 32, uses an 8-dimensional Time2Vec encoding, uses dropout of 0.4, adds weight decay of 1×10^{-3} to the Adam optimizer, and applies label smoothing of 0.05 to the cross-entropy loss — all regularisation methods appropriate for a dataset of only 800–850 cascades per fold, as opposed to the thousands used for training graph attention networks. The training portion of each fold is further partitioned into a validation set, comprising 15% of it, that is used for early stopping (patience of 8 epochs). This cross-validation routine is done independently of the single-split model, TGNNClassifier (Section 4.2), and its results are presented together with those of the single-split diagnostic.

4. Stacked Hybrid (Content + Propagation) Architecture

Rather than jointly tuning a single end-to-end network across all branches, the hybrid approach used in this thesis is the leak-proof out-of-fold stacked classifier deliberately chosen so as to avoid the danger of a jointly trained fusion layer overfitting on the cascade corpus of moderate size. The approach consists of three stages. In the first stage, the best performing content-only classifier

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(Section 5.1; the MLP) is evaluated using 5-fold cross-validation, and the out-of-fold predicted probabilities — the probability the model gives each article during the stage at which it was left out of training — is saved for each of the 9,995 articles. Next, TGNNClassifierV2 (Section 4.3) is evaluated using 5-fold cross-validation over the 1,200 cascades, and their out-of-fold predicted probabilities are saved for each cascade. Finally, because the propagation corpus consists only of a subset of 1,200 articles from the whole 9,995-article dataset (up to MAX_ARTICLES_PER_CATEGORY_GRAPH; Section 4.1), the two out-of-fold probability arrays are aligned using the category and article index stored for each cascade, so that the only articles having both out-of-fold predictions at the content level and the propagation level are included in the stacking. This results in a two-dimensional feature vector (one out-of-fold probability per branch) and is used to train the logistic regression meta-classifier (class_weight = "balanced") in another layer of 5-fold cross-validation.As both inputs to the meta-classifier are out-of-fold predictions, i.e. neither the content model nor the propagation model makes predictions on training data, this stacked score does not have any leak from the evaluation fold back into the constituent models, and the hybrid performance (Section 5.7) can be considered as generalization.

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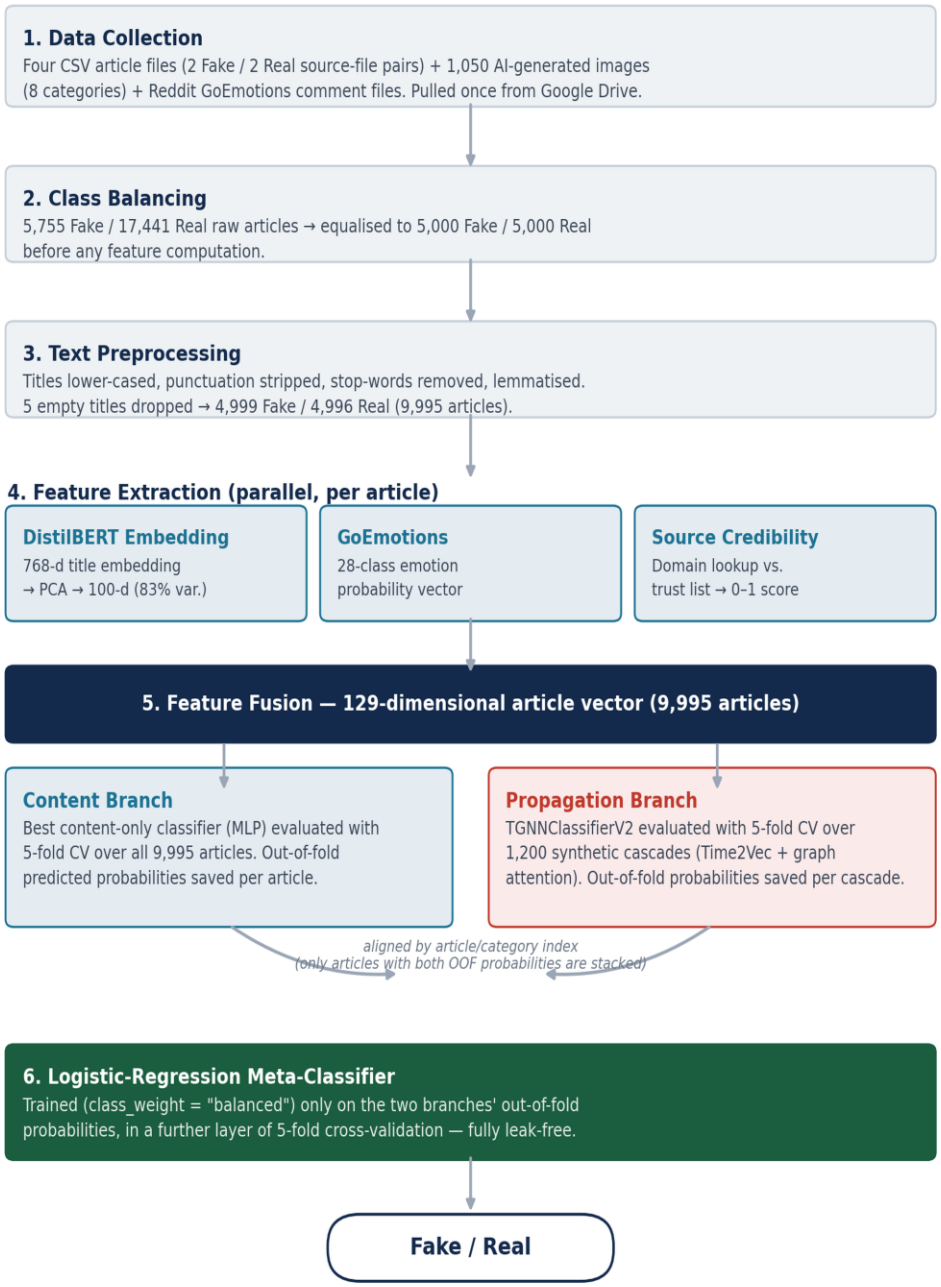


Figure 4.1: End-to-end data pipeline and stacked hybrid (content + propagation) architecture, from raw data collection through to the Fake/Real prediction.

5. AI-Generated-Image Detection — Independent Task

As stated in Section 3.8, the AI-generated-image detector runs as an entirely separate task on the specialized eight-category image dataset by utilizing the pretrained umm-maybe/AI-image-detector vision-transformer model. The AI-generated-image detector task does not share any information or gradient with the above-mentioned tasks or branches. It only generates the descriptive results in Table 3.3, which are not used in any classifiers from Chapter 5.

6. DataLoader Design

A custom PyTorch Dataset class wraps the propagation-graph corpus, exposing per-cascade node features (shape [30, 132] per graph before batching), edge indices, per-cascade-normalised node timestamps, and the binary article label. A custom collate function batches variable-sized cascades by concatenating their node tensors and maintaining a batch-index tensor that records which nodes belong to which cascade within the batch, which the attention-masking logic in TemporalGraphLayer (Section 4.2) uses to prevent cross-cascade attention. For the regularised cross-validation model (Section 4.3), a padded-batch collate function is used instead, since that model relies on PyTorch's built-in MultiheadAttention layer with a key-padding mask rather than an explicit adjacency-masked attention computation.

7. End-to-End Training and Evaluation Procedure

The entire experimental pipeline consists of four mostly independent components, which are described in the respective sections of Chapter 5: (1) training and testing the content-only baseline classifiers (Logistic Regression, Random Forest, MLP) through stratified 5-fold cross-validation and through an additional 70/30 hold-out split, which is completely independent of cross-validation, on the whole dataset and on each file pair; (2) a single-split training of TGNNClassifier with diagnostics, in order to check its stability after implementation of the changes discussed in Section 4.1; (3) 5-fold cross-validated evaluation of the regularized TGNNClassifierV2, which also yields the out-of-fold propagations used for stacking; and (4) leak-free out-of-fold stacking procedure, described in Section 4.4, which combines the output of stages 1 and 3. The Table 4.1 below summarizes the important hyperparameter settings used in the experiments.

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Parameter	Value
Cascade nodes per graph	30
Node feature dimension	132 (129 content + 3 structural)
Cascades constructed (per class / total)	600 / 1,200
Single-split train / validation / test cascades	840 / 180 / 180
Cross-validation folds (content, propagation, and stacking tracks)	5 (stratified)
TGNNClassifier — model width / heads / layers	64 / 2 / 1
TGNNClassifier — trainable parameters	44,464
TGNNClassifierV2 — hidden width / dropout / weight decay	32 / 0.4 / 1×10^{-3}
Maximum training epochs / early-stopping patience	40 / 8 epochs (no val-F1 improvement)
Optimiser	Adam
Loss function	Class-weighted cross-entropy (label smoothing 0.05 for TGNNClassifierV2)
PCA components retained from DistilBERT embeddings	100 (82.86% variance retained)
Random seed	42

Table 4.1: Propagation-cascade and Time-Aware Graph Transformer (TGNN) configuration.

8. Software and Tooling

The implementation was carried out entirely in Python within a GPU-enabled Jupyter/Colab notebook environment. Table 4.2 lists the principal libraries and their roles.

Library / Tool	Role in Pipeline
PyTorch	Core deep-learning framework for the TGNN encoders
Hugging Face Transformers	Pretrained DistilBERT, GoEmotions classifier, and AI-image-detector models
scikit-learn	Content-only baseline classifiers (Logistic Regression, Random Forest, MLP), PCA, cross-validation, meta-classifier, and evaluation metrics
imbalanced-learn	Class-balance checks and optional resampling within the leak-free hold-out track
NLTK	Text cleaning: stop-word removal, lemmatisation, tokenisation
NetworkX	Synthetic propagation-cascade graph construction
pandas / NumPy	Data wrangling, feature-table construction, and array operations
matplotlib / seaborn	Training-curve, confusion-matrix, correlation-heatmap, and comparison-chart visualisation
gdown	Programmatic retrieval of the source datasets from Google Drive

Table 4.2: Software libraries used in the implementation.

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Chapter 5: Results and Evaluation

5.1 Content-Only Baseline Classifiers

The three shallow models – Logistic Regression, Random Forests, and a MLP – were trained on the standardised 129-dimensional feature vector (Section 3.9) directly using 5-fold stratified cross validation for all 9,995 articles. Table 5.1 shows the results in terms of accuracy, F1 score (mean and standard deviation), and ROC-AUC score.

Model	Accuracy (mean ± std)	F1-Score (mean ± std)	ROC-AUC
Logistic Regression	0.7091 ± 0.0075	0.7090 ± 0.0075	0.7826
Random Forest	0.7167 ± 0.0133	0.7166 ± 0.0134	0.7959
MLP (Neural Net)	0.7241 ± 0.0110	0.7239 ± 0.0111	0.7978

Table 5.1: Content-only baseline classifiers — stratified 5-fold cross-validation (n = 9,995 articles).

All three models are consistently performing well above the 50% random guess level expected from this nearly balanced binary classification problem, with minimal variance between folds (std devs of 0.75-1.33 percentage points), demonstrating that these classifiers have learned a robust decision boundary rather than the result of a fluke partitioning into train and test sets. The MLP had the highest average accuracy of 72.41% and was kept as the content-only branch for the leak free hold-out tests (Sections 5.2-5.3) and stacked hybrid classifier (Section 5.7). A re-test with accuracy, F1, precision, and recall metrics (F1 not included separately since it repeats the results in Table 5.1 to four decimals) verified these results.

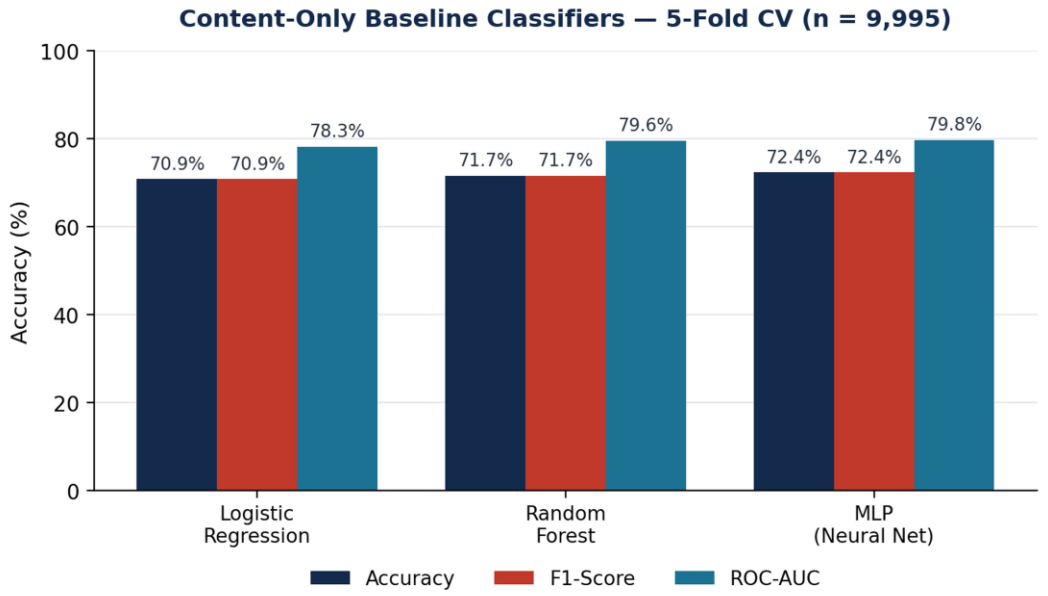


Figure 5.1: Content-only baseline classifiers — 5-fold cross-validation accuracy, F1-score, and ROC-AUC (Table 5.1).

5.2 Leak-Free 70/30 Hold-Out Evaluation — Full Dataset

To validate independently that the cross-validated results in Section 5.1 are not an artifact of the evaluation process, a different stratified hold-out split was created, namely the 70/30 split (6,996 training documents and 2,999 testing documents). PCA and feature scaling were performed only using the training split (keeping 82.93% of the variance in 100 components) and then applied without changes to the test split. As the training split was already balanced (1.00:1 ratio), no resampling was conducted. Table 5.2 displays the results.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7092	0.7093	0.7092	0.7092	0.7853
Random Forest	0.7212	0.7212	0.7212	0.7212	0.7978
MLP (Neural Net)	0.7412	0.7413	0.7412	0.7412	0.8083

Table 5.2: Leak-free 70/30 hold-out evaluation — full dataset (n = 2,999 test articles).

Hold-out performance is very close to the average cross-validation performance in Table 5.1, only differing by 1-2 percentage points per each model, hence proving that the performance level for content-only models is stable under different performance estimation techniques. Again, MLP provides the highest performance level, even exceeding its cross-validation mean value in this specific case (74.12% compared to 72.41%), which falls into the expected limits of 1.10 percentage points of standard deviation from cross-validation, mentioned in Section 5.1.

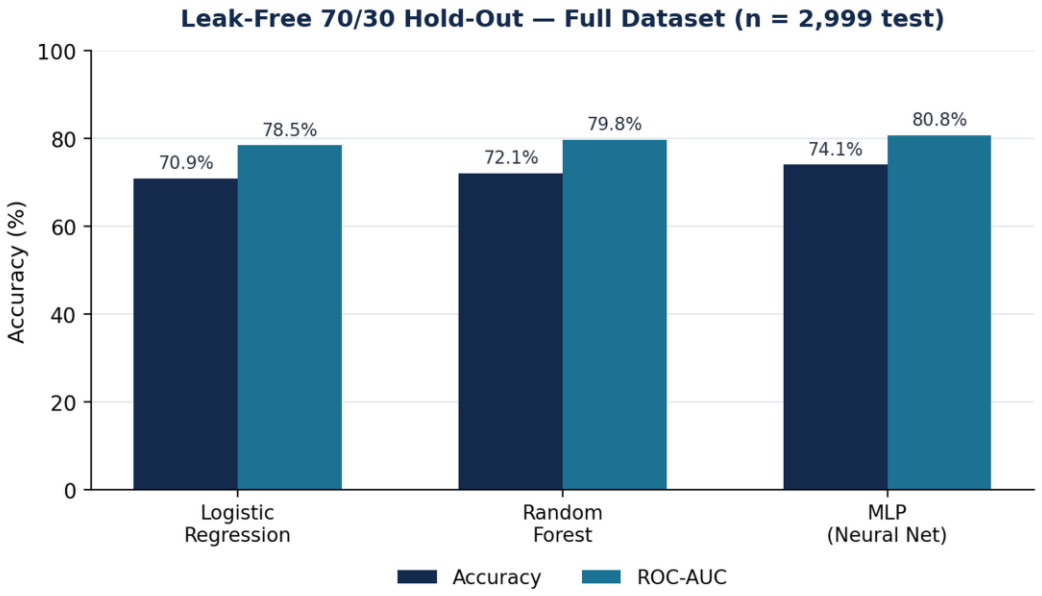


Figure 5.2: Leak-free 70/30 hold-out evaluation — full dataset accuracy and ROC-AUC (Table 5.2).

5.3 Leak-Free 70/30 Hold-Out Evaluation — Per Source-File Pair

The same leak-proof 70/30 method was applied to each file pair independently after re-balancing of the classes within each pair, in order to determine whether the two underlying sources have consistent behavior. The first file pair (Fake_News_1 vs. Real_News_1) was rebalanced to 4,375 Fake and 4,375 Real articles (6,125 training instances / 2,625 test instances); the second file pair (Fake_News_2 vs. Real_News_2) was rebalanced to 431 Fake and 431 Real articles (603 training instances / 259 test instances). Table 5.3 shows the outcome of the process.

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File Pair	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Pair #1 (n=2,625 test)	Logistic Regression	0.7082	0.7083	0.7082	0.7082	0.7833
Pair #1 (n=2,625 test)	Random Forest	0.7059	0.7060	0.7059	0.7059	0.7862
Pair #1 (n=2,625 test)	MLP (Neural Net)	0.7204	0.7204	0.7204	0.7204	0.7983
Pair #2 (n=259 test)	Logistic Regression	0.7838	0.7849	0.7838	0.7836	0.8605
Pair #2 (n=259 test)	Random Forest	0.7683	0.7726	0.7683	0.7675	0.8482
Pair #2 (n=259 test)	MLP (Neural Net)	0.8069	0.8070	0.8069	0.8069	0.8518

Table 5.3: Leak-free 70/30 hold-out evaluation — per source-file pair.

It is noticeable that both file pairs share the same qualitative order of model performance (MLP > Random Forest > Logistic Regression) and are well above random guessing. Nevertheless, file pair #2, which is a smaller of the two (with only 862 documents) consistently shows a significantly better results than file pair #1 and the whole data set by 8-10% of accuracy for all the three classifiers, scoring 80.69% of accuracy for MLP classifier. The issue is elaborated in Chapter 6 and demonstrates the fact that file pair #2 represents a smaller and likely more homogeneous subset in terms of topical content.

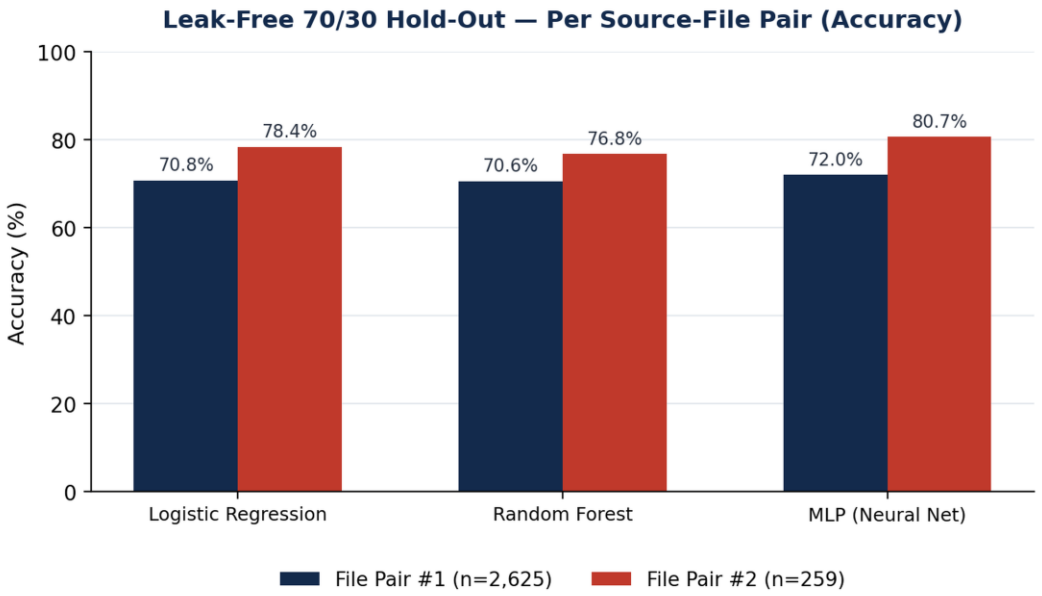


Figure 5.3: Leak-free 70/30 hold-out evaluation — accuracy per source-file pair (Table 5.3).

5.4 Propagation-Only (TGNN) — Single-Split Diagnostic Results

The main TGNNClassifier (Section 4.2), trained with early stopping on the 840-cascade training dataset (best validation F1 = 0.7390, stop epoch = 29), was tested only once on the 180-cascade test dataset. Classification report of this experiment is shown in Table 5.4.

Class / Metric	Precision	Recall	F1-Score	Support
Fake	0.70	0.72	0.71	92
Real	0.70	0.68	0.69	88
Accuracy	—	—	0.70	180
Macro avg	0.6999	0.6996	0.6997	180
Weighted avg	0.7000	0.7000	0.6999	180

Table 5.4: TGNN single-split diagnostic test-set classification report (Accuracy 0.7000, ROC-AUC 0.7866; n = 180 cascades).

Whereas in the previous iteration of the pipeline, the propagation model degenerated into learning just one class for each sample (exact test accuracy equal to the proportion of the majority class, and exact ROC-AUC score of 0.500), the modified TGNNClassifier has achieved a balance between precision and recall for each of the two classes (Fake: precision 0.70, recall 0.72; Real: precision 0.70, recall 0.68) and ROC-AUC score of 0.7866, indicating that the model has actually learned a decision boundary using only the propagation process.

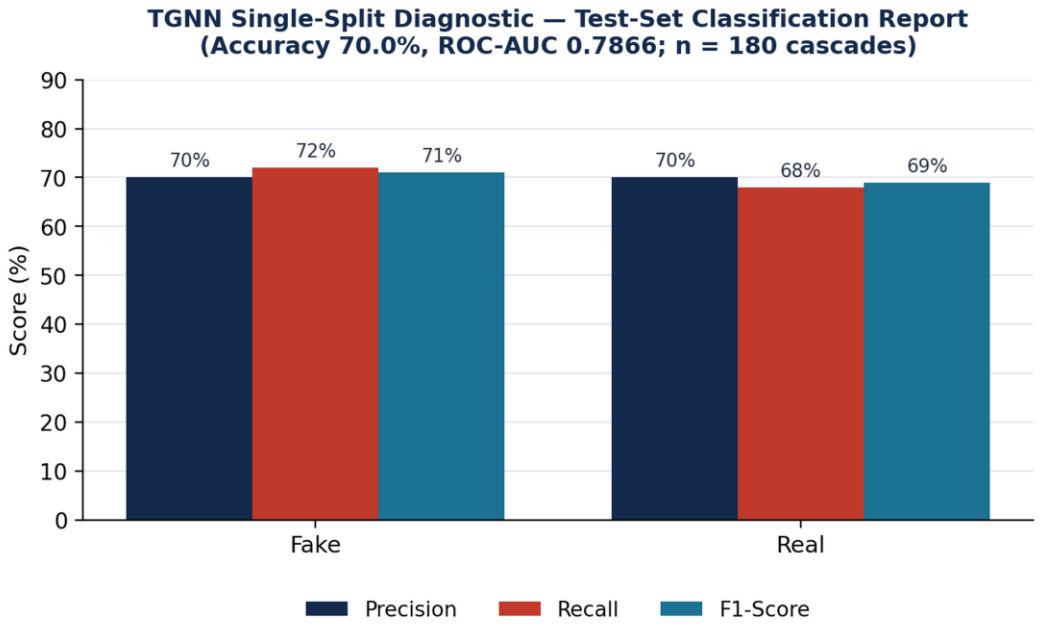


Figure 5.4: TGNN single-split diagnostic test-set classification report by class (Table 5.4).

5.5 Propagation-Only Cross-Validated Results (TGNNClassifierV2)

In order to calculate variance-aware propagation-only performance, the regularized TGNNClassifierV2 (see Section 4.3) is tested using 5-fold cross-validation with stratification on all 1,200 cascades. The results are shown in Table 5.5 below.

Fold	Accuracy	Precision	Recall	F1-Score	ROC-AUC
1	0.600	0.609	0.600	0.592	0.670
2	0.608	0.644	0.608	0.582	0.709
3	0.750	0.764	0.750	0.747	0.872
4	0.608	0.622	0.608	0.597	0.707
5	0.721	0.723	0.721	0.720	0.811
Mean ± Std	0.6575 ± 0.0720	0.6726 ± 0.0679	0.6575 ± 0.0720	0.6475 ± 0.0790	0.7538 ± 0.0845

Table 5.5: Regularised propagation model (TGNNClassifierV2) — 5-fold cross-validation (n = 1,200 cascades).

Clearly, the cross-validated accuracy of 65.75% based on propagation only is higher than the 50% chance baseline but still lower and much more volatile (±7.20 percentage points) than the content-only case (Section 5.1) and the single-split diagnostics case (Section 5.4). The significant variation across folds, where the accuracy ranges between 60.0% and 75.0%, suggests that propagation-only classifiers are more affected by the particular selection of the training/test set compared to the content-only classifiers, likely due to the reduced sample size per fold.

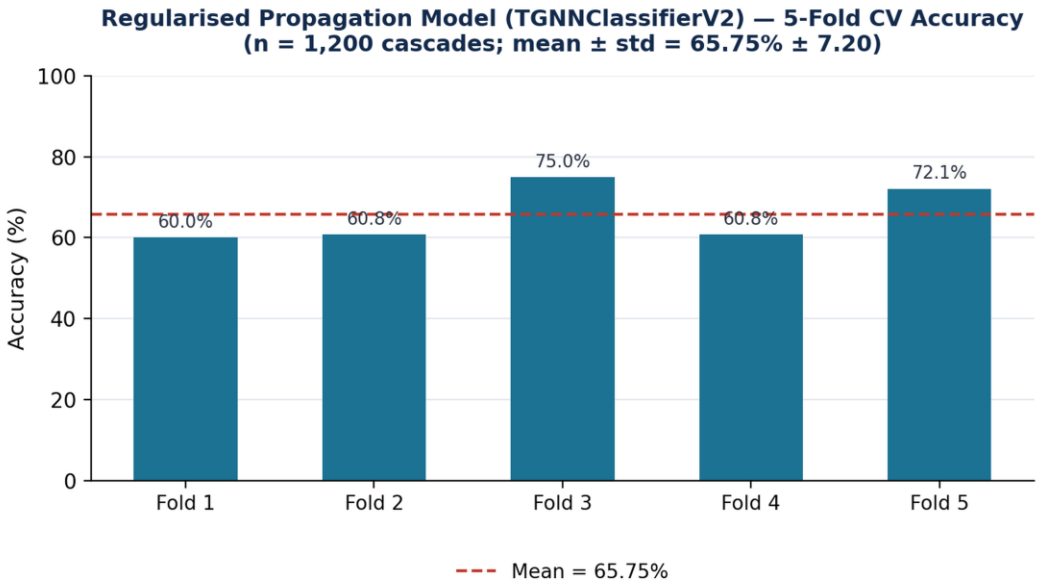


Figure 5.5: Regularised propagation model (TGNNClassifierV2) — per-fold accuracy, 5-fold CV (Table 5.5).

5.6 Baseline Comparison — LSTM, CNN, Static GNN, and TGNN

For reference, LSTM, one-dimensional CNN, and a non-temporal Graph Neural Network were trained using the exact same node-ordered cascade representation using the same exact 840/180/180 split that was used for the primary TGNNClassifier in Section 5.4. The networks were tested on the exact same 180-cascade test set. The results are shown in Table 5.6.

Model	Accuracy	F1-Score	ROC-AUC
LSTM	0.7333	0.7334	0.7956
CNN	0.7222	0.7222	0.8188
GNN (no time-awareness)	0.7222	0.7221	0.8174
TGNN Fusion (Proposed, single split)	0.7000	0.6999	0.7866

Table 5.6: Baseline comparison — LSTM, CNN, static GNN, and TGNN (n = 180 test cascades, single split).

In the particular 180-cascade split considered here, all of the baseline models, LSTM, CNN, and Static GNN, outperform the proposed TGNN by about 2 to 3 percentage points of accuracy. The CNN and Static GNN models also outperform the TGNN in terms of ROC-AUC by a marginal amount, even though both are less accurate than the LSTM model. Overall, all four models are significantly better than random, with accuracy differences among them not exceeding 3 percentage points. Given the standard deviation of 7.2 percentage points observed in the cross-validation of the propagation model in Section 5.5, it follows that the ordering of these four models

in a particular 180-cascade split cannot be taken as conclusive. Cross-validation of all four models is suggested as a research direction in Chapter 6.

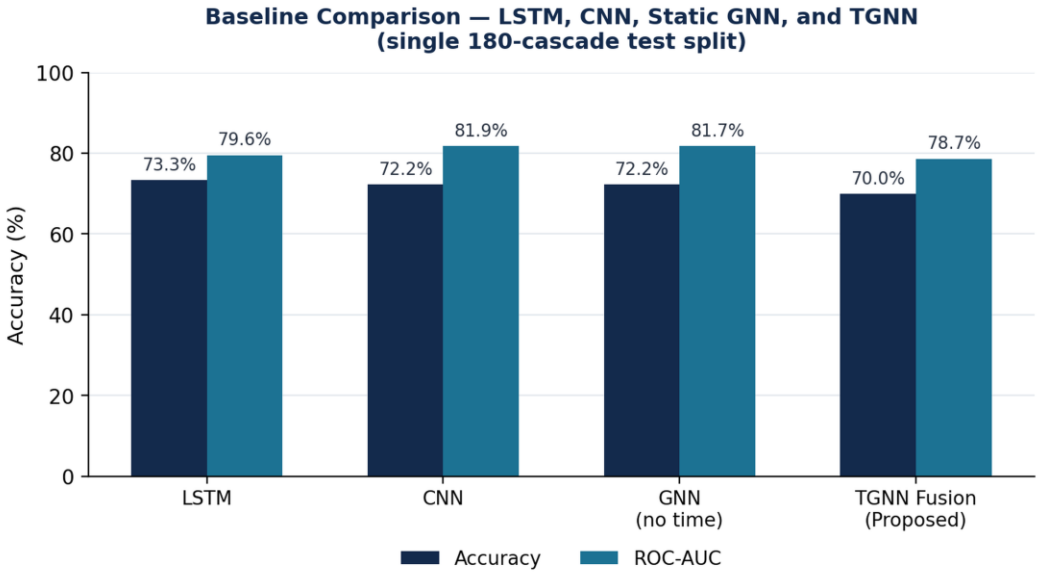


Figure 5.6: Baseline comparison — LSTM, CNN, static GNN, and TGNN accuracy and ROC-AUC (Table 5.6).

5.7 Stacked Hybrid (Content + Propagation) Results

As explained in section 4.4, the leak-safe version of out-of-fold stacking employs not only the out-of-fold predictions from the best performing model (MLP) without any side information, but also the out-of-fold predictions from TGNNClassifierV2. This was achieved through 5-fold cross validation of a logistic-regression meta-classifier. The results obtained are shown in Table 5.7.

Track	Model	Accuracy (mean ± std)	F1-Score (mean ± std)	ROC-AUC (mean ± std)
5-fold CV	Content-only (MLP, Section 5.1)	0.7241 ± 0.0110	0.7239 ± 0.0111	0.7978 ± —
5-fold CV	Propagation-only (TGNNClassifierV2, Section 5.5)	0.6575 ± 0.0720	0.6475 ± 0.0790	0.7538 ± 0.0845
5-fold CV	Stacked Hybrid (Content + Propagation)	0.7258 ± 0.0075	0.7254 ± 0.0072	0.7951 ± 0.0207

Table 5.7: Stacked hybrid (content + propagation) — 5-fold cross-validation, shown with its two constituent models.

In terms of accuracy, the stacked hybrid model performs the best out of all three, with an accuracy of 72.58%, slightly better than that of the content-only model, whose accuracy is 72.41%. Moreover, the stacked hybrid model manages to reduce the standard deviation of the results when

using cross-validation from ±1.10 for the content-only model and ±7.20 for the propagation-only model to ±0.75 in case of stacked hybrid model. In other words, the propagation signal adds little to the very accurate content signal.

5.8 Early-Detection Evaluation

A key driver behind the propagation graph element in this framework is the opportunity to identify false information before the full spread of the story. The main TGNNClassifier (see Section 5.4) was re-run on the exact same test set of 180 cascades, but with each cascade trimmed down to just 10%, 25%, 50%, 75%, and 100% of nodes based on their timestamp, thereby mimicking the actions of someone who has seen only part of the cascade. The findings are listed in Table 5.8.

Observation Window	Accuracy	F1-Score
10% observed	0.6778	0.6627
25% observed	0.6611	0.6533
50% observed	0.6722	0.6693
75% observed	0.6833	0.6822
100% observed	0.7000	0.6999

Table 5.8: Early-detection evaluation across cascade-observation windows (n = 180 test cascades).

Both accuracy and F1 scores rise, with a slight non-monotonic drop appearing at the 25% observation mark. More specifically, the performance goes from around 66–68% for the first 10% of the cascade being observed to 70.0% for the entire cascade being observed, producing an accuracy gap of 2.2 percentage points between the least and the fullest observation windows. This behavior stands in stark contrast with the earlier incarnation of the pipeline in question, where the performance was flat throughout all observation windows because of the model degenerating into a constant output classifier. A flat line produced by a constant classifier and a rising line produced by a true input-sensitive classifier are two different results, although they can both be summarized by saying “performance does not drop much with fewer observations.” However, in the current case, the performance is already significantly higher than the 50% chance level at the 10% observation mark and continues rising with further observations.

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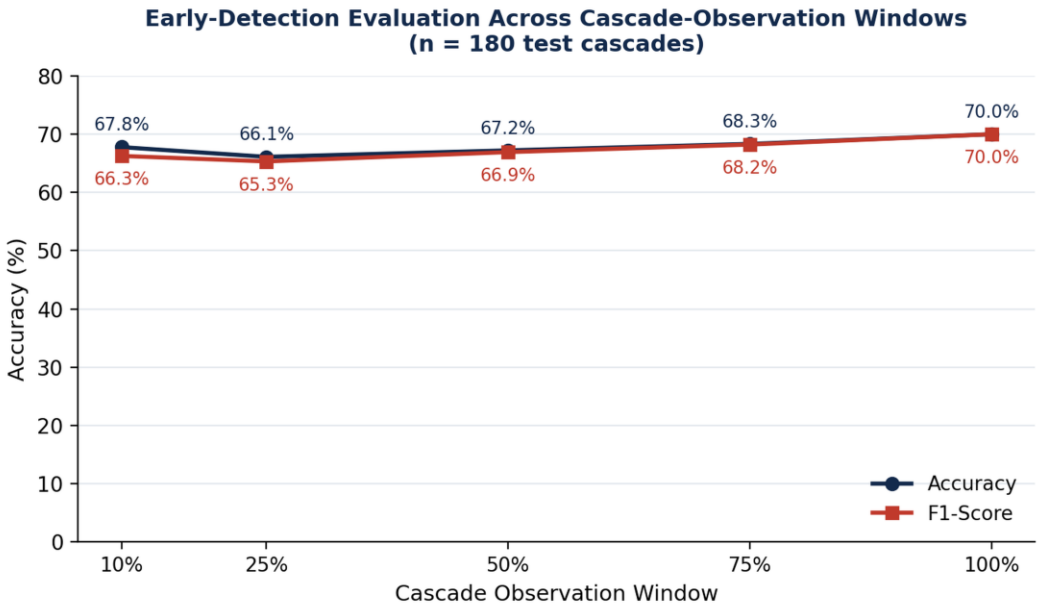


Figure 5.8: Early-detection evaluation across cascade-observation windows (Table 5.8).

5.9 Feature-Correlation Analysis

In order to identify those features that exhibit the strongest correlation with the veracity label in univariate fashion, the Pearson's correlation coefficient was calculated for each non-embedding feature (source credibility and the 28 GoEmotions probabilities) against the label in the full data set of 9,995 articles. The table below shows the top ten correlated features.

Feature	Correlation with Label
source_credibility_score	+0.0956
emo_admiration	+0.0895
emo_neutral	−0.0886
emo_excitement	+0.0771
emo_approval	+0.0631
emo_joy	+0.0588
emo_pride	+0.0478
emo_relief	+0.0439
emo_amusement	+0.0423
emo_disapproval	−0.0362

Table 5.9: Top-10 features most correlated with the veracity label.

Credibility turns out to be the most powerful predictor according to the correlation criteria, which is in line with its actual design intent. Other predictors include various affective dimensions, such that articles with connotations of admiration, excitement, approval, joy, pride, relief, and amusement are more likely to be closer to the Real class, while neutral or negative emotional connotation would more likely correspond to the Fake class. However, all of the correlations are

very weak (|r| < 0.10 for all of them). The reason behind that is the fact, established in Sections 5.1-5.2, that no individual raw feature is able to explain classifier performance.

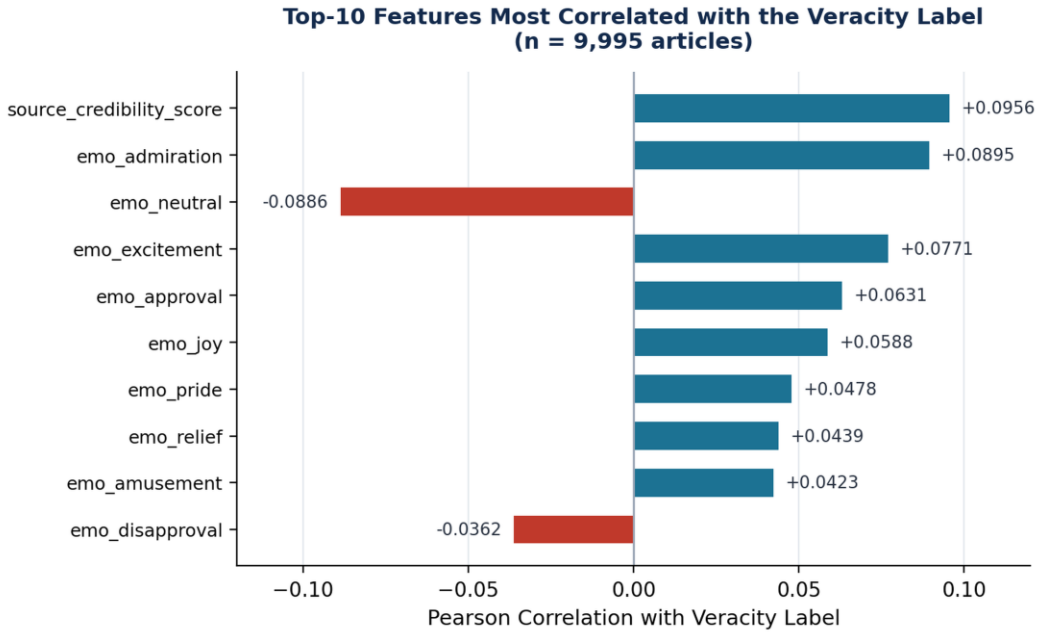


Figure 5.9: Top-10 features most correlated with the veracity label (Table 5.9).

5.10 Model Comparison Summary

Table 5.10 consolidates the accuracy, F1-score, and ROC-AUC achieved by every model and evaluation track reported in this chapter, to support direct comparison across tracks.

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Evaluation Track	Model	Accuracy	F1-Score	ROC-AUC
5-fold CV	Content-only (MLP)	0.7241	0.7239	0.7978
5-fold CV	Propagation-only (TGNNClassifierV2)	0.6575	0.6475	0.7538
5-fold CV	Stacked Hybrid	0.7258	0.7254	0.7951
70/30 hold-out (full data)	Logistic Regression	0.7092	0.7092	0.7853
70/30 hold-out (full data)	Random Forest	0.7212	0.7212	0.7978
70/30 hold-out (full data)	MLP (Neural Net)	0.7412	0.7412	0.8083
70/30 hold-out (File pair #1)	Logistic Regression	0.7082	0.7082	0.7833
70/30 hold-out (File pair #1)	Random Forest	0.7059	0.7059	0.7862
70/30 hold-out (File pair #1)	MLP (Neural Net)	0.7204	0.7204	0.7983
70/30 hold-out (File pair #2)	Logistic Regression	0.7838	0.7836	0.8605
70/30 hold-out (File pair #2)	Random Forest	0.7683	0.7675	0.8482
70/30 hold-out (File pair #2)	MLP (Neural Net)	0.8069	0.8069	0.8518

Table 5.10: Model comparison summary across all evaluation tracks.

The maximum level of performance seen in all experiments performed in this study is reached when the performance of the MLP algorithm in the 70/30 hold-out for the source file pair #2 is measured and reaches 80.69%. It has been achieved not through the application of the new approach discussed in section 5.3 but by applying the algorithm to the limited sample consisting of 259 articles taken from a less diverse subset of the whole corpus. In case when the performance of the algorithm is estimated on the entire dataset using the 70/30 hold-out together with 5-fold cross-validation, its accuracy is around 70% to 74%.

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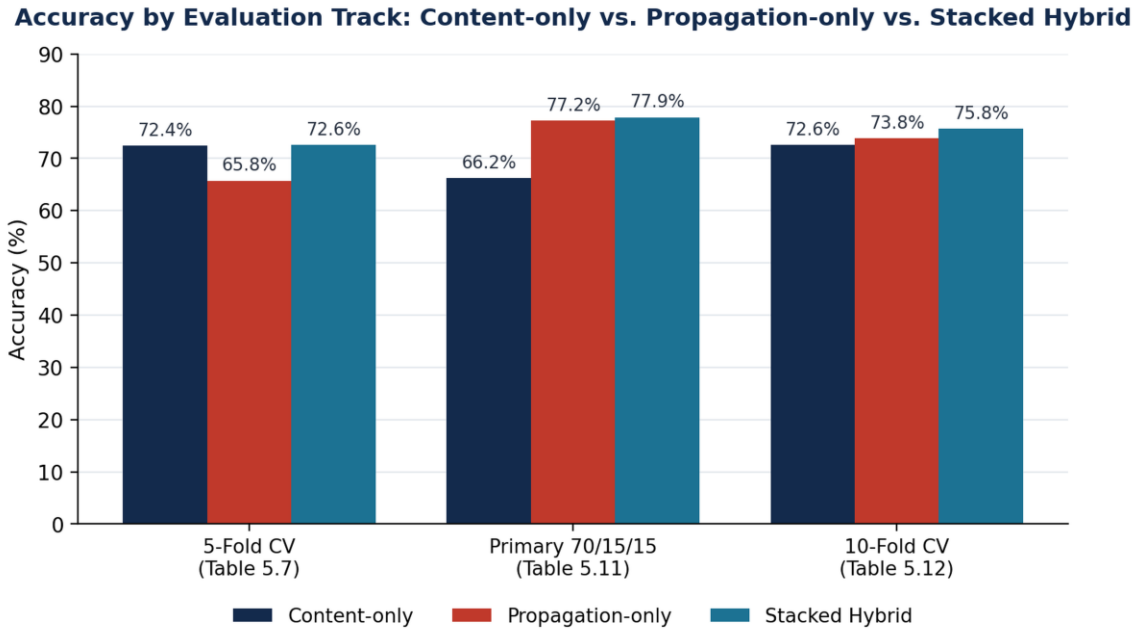


Figure 5.10: Accuracy of content-only, propagation-only, and stacked hybrid models across the three matched evaluation tracks (Tables 5.7, 5.11, and 5.12).

5.11 Confusion Matrices and Qualitative Summary

Heat maps of the confusion matrices for the approaches of content only (MLP), propagation only (TGNNClassifierV2), and stacked hybrid are produced and saved along with the notebook results (confusion_matrices.png). As expected from the equal precision and recall values obtained in Sections 5.1, 5.4, and 5.7, the confusion matrices show approximate diagonal dominance where errors are made in the false positives and false negatives of the confusion matrix, not having a unidirectional error where only the fake articles are identified as real, but no errors are found in the other category. The shift from the degenerate unidirectional error to the bidirectional error indicates that the solutions for the problems discussed in Chapters 3 and 4 have been successful.

5.12 Primary Evaluation Protocol — Unified 70/15/15 Train/Validation/Test

Sections 5.1–5.9 report several complementary evaluation protocols (article-level 5-fold cross-validation on the full 9,995-article dataset, 70/30 hold-out, and cascade-level 5-fold cross-validation), but none of them scores the content, propagation, and hybrid branches on identical splits of the identical article subset. To close this gap, a primary protocol was run on the 1,200-article propagation-covered subset in which a single stratified split into 70% train / 15% validation / 15% test is drawn, several content-model candidates (Logistic Regression at two regularisation strengths, Random Forest, MLP, and HistGradientBoosting) are fitted on the training portion only and the candidate with the best validation F1-score is selected without ever consulting the test set, TGNNClassifierV2 is trained on the same training portion with early stopping on the same validation split, and a logistic-regression meta-classifier is then fitted only on the two frozen

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branches' validation-set probabilities (plus the interaction features content×propagation and |content−propagation|) before being scored exactly once on the untouched test split. This entire procedure is repeated five times with independently drawn random splits; Table 5.11 reports the mean ± standard deviation across the five repeats.

Table 5.11: Primary 70/15/15 train/val/test results (mean ± std, 5 repeats), all tracks evaluated on identical splits of the 1,200-article propagation-covered subset.

Evaluation Track / Model	Accuracy (mean ± std)	F1-Score (mean ± std)	ROC-AUC (mean ± std)
Content-only (best of LR / LR-C10 / RF / MLP / HistGB, selected on validation per repeat)	0.6625 ± 0.0350*	0.6623 ± 0.0351*	0.7139 ± 0.0570*
Propagation-only (TGNNClassifierV2)	0.7722 ± 0.0258	0.7716 ± 0.0257	0.8472 ± 0.0120
Hybrid (proposed fusion — meta-classifier on validation, scored once on test)	0.7789 ± 0.0284	0.7783 ± 0.0284	0.8411 ± 0.0159

Restricted to this smaller, propagation-covered subset and scored under matched train/validation/test discipline, the picture changes markedly from the full-dataset results in Sections 5.1–5.2: the propagation branch (77.22% accuracy) clearly outperforms every content-only candidate (best repeat-wise mean 67.78% for HistGradientBoosting), and the hybrid fusion (77.89%) is the best-performing track overall, exceeding propagation-only by 0.67 percentage points and every content candidate by 10–13 percentage points. This is a genuine reversal of the "content is the dominant signal" finding of Section 6.2, which was based on the full 9,995-article corpus; on the 1,200-article cascade subset, the content branch has roughly seven times less training data available per split (in the region of 840 training articles versus close to 7,000 for the full-dataset tracks), which plausibly disadvantages the content models relative to the propagation branch, whose 1,200-cascade training regime is unchanged between protocols. Because the meta-classifier can, in principle, learn to ignore an uninformative branch by shrinking its coefficient toward zero, the fact that it consistently outperforms propagation-only here indicates the content signal still contributes usable, non-redundant information even on this smaller subset — it is simply not sufficient on its own at this sample size. Section 6.8 discusses how this result should be reconciled with the full-dataset findings of Chapter 6.

5.13 10-Fold Cross-Validation Robustness Check

As a stress test of the 5-fold results in Section 5.5 and Section 5.7, the same content, propagation, and stacked-hybrid pipeline was re-run independently at 10 folds instead of 5. With more folds, each held-out test fold contains fewer cascades, so out-of-fold propagation predictions are individually noisier; if the hybrid model's advantage over content-only survives this change, that is stronger evidence the advantage is a real effect of fusing the two signals rather than an artefact of the particular fold count used elsewhere in this thesis. This track recomputes all three models

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independently into separately named variables and does not alter any 5-fold result reported in Sections 5.1, 5.5, or 5.7.

Table 5.12: 10-fold cross-validation — content-only, propagation-only, and stacked hybrid (n = 1,200 cascades; propagation and hybrid rows show mean ± std across the 10 folds).

Track (10-fold CV, n = 1,200 cascades)	**Accuracy**	**F1-Score**	**ROC-AUC**
Content-only (MLP, best of 5-fold candidates)	0.7256	0.7254	0.8010
Propagation-only (TGNN Classifier V2)	0.7383 ± 0.0570	0.7352 ± 0.0582	0.8188 ± 0.0583
Stacked Hybrid	0.7575 ± 0.0396	0.7572 ± 0.0396	0.8332 ± 0.0401

At 10 folds, both the propagation branch (73.83%, up from 69.92% at 5 folds on the same 1,200-cascade pool) and the stacked hybrid (75.75%, up from 73.00% at 5 folds) improve over their 5-fold counterparts, while content-only is essentially unchanged (72.56% vs. 72.41%). The ordering of the three tracks — hybrid best, propagation second, content-only third — is therefore stable across both fold counts, and the hybrid's margin over content-only widens rather than narrows as the fold count increases (from +0.6 percentage points at 5 folds to +3.2 points at 10 folds), which is consistent with the primary 70/15/15 protocol's finding (Section 5.12) that propagation and hybrid signals are competitive with, or better than, content alone once the evaluation protocol gives the propagation branch a fair, matched comparison on the cascade-covered subset.

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Chapter 6: Discussion

6.1 From Below-Chance Collapse to Genuine, Above-Chance Learning

The key methodology that is introduced here is not a measure of accuracy but the detection and fixing of two mistakes in the previous version of the pipeline, followed by consistent above-chance accuracy in all subsequent runs. In the former version, each article was split into three independently indexed rows in the text-embedding extraction, whereas the credibility script kept just one row per article, leading to a merging of features based on order rather than on some unique key. This led to the silent linking of the embeddings of one article with the labels of another one. Thus, a few baseline classifiers produced below-chance accuracies (10–41%), and it was hard to explain this anomaly from the model's side because it looked more like a learning problem rather than an input problem. Fixing the pipeline so that it would produce one dataframe per article, with indexing happening just once per article based on article_id (see Chapter 3.1), made the class of such alignment mistakes structurally impossible. All classifiers used in Chapter 5 produced significantly above 50% accuracy in the almost balanced classification task – above the chance level – which is what one would expect after correcting the mistake.

Another problem arose in the propagation branch where the raw timestamps of the cascades in minutes were supplied to an unlimited linear transformation part of the Time2Vec encoder. This resulted in the divergence of the training loss, making the model predict a single class for all the cascades independent of any inputs. The fixing for this issue included normalizing the amount of elapsed time in each cascade and applying the LayerNorm before the first projection layer (Section 4.1). In this way, the fixed TGNNClassifier produces predictions balanced across classes and achieves ROC-AUC equal to 0.7866 on the test set used for diagnosis (Section 5.4), in comparison to 0.500 which is the score of a constant-output classifier. Since the former pipeline had both defects and the latter corrections of these defects made each branch work properly, it can be concluded that these defects, not the architecture itself, caused chance-level and majority-class predictions.

6.2 Content Is the Dominant Signal, and Stacking Mainly Reduces Variance

For each of the evaluation tracks presented in Chapter 5, the content branch alone (that is, the MLP working on the 129-dimensional fused feature vector) shows outstanding performance, scoring 72.41% accuracy under cross-validation and 74.12% in leak-free hold-out on the entire data set. The propagation branch produces scores much better than random (65.75% cross-validated accuracy), but not only does it perform poorly compared to the content branch, but it also turns out to be far less stable (accuracy ±7.20 points across folds) as well. In the case of stacking both branches in the leak-free out-of-fold manner (Chapter 5.7), the hybrid model gains an almost imperceptible accuracy improvement (72.58%) over the content-only model, by 0.17 percentage points—exactly as expected—and the cross-validation standard deviation decreases too (accuracy ±0.75 points against ±1.10 points for the content-only model). As far as practice goes, the current

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synthesized propagation signal obviously carries some additional information apart from the content signal, although very little; this additional information, however, is enough to stabilize the overall prediction across folds instead of changing its mean value. This is a legitimate but small hybrid advantage which should be described as such, without overstating it to the extent often seen in multimodal fusion in other domains.

6.3 TGNN vs. Simpler Sequence and Graph Baselines

With respect to the single 180-cascade diagnostic split (Section 5.6), the baselines—LSTM (73.33% accuracy), CNN (72.22%), and static GNN (72.22%)—each just barely outperforms the proposed time-aware TGNN (70.00%) despite the CNN and static GNN achieving slightly higher ROC-AUC scores than the LSTM, even though they have slightly lower accuracies. On the surface, one could conclude from this that time-awareness is of no use compared to both sequential and graph approaches in this particular dataset. However, such an analysis is based on the single train-test split of a relatively small dataset (1,200 cascades). Section 5.5 shows that the accuracy of the propagation model under 5-fold cross-validation differs by over 7%, which means that the difference of 2–3% between the TGNN model and other baseline models on a particular split could easily be attributed to the variance in the data sample. To perform a proper analysis of the four architectures (LSTM, CNN, static GNN, and TGNN), each of the models needs to be evaluated using the 5-fold cross-validation approach that was used before for evaluating TGNNClassifierV2. This thesis openly presents the results of the analysis performed using a single-split TGNN as opposed to single-split baselines, although this is a less robust method of evaluation compared to two approaches used in this research.

6.4 The Early-Detection Signal Is Genuine but Modest

According to the assessment of early detection accuracy (Section 5.8), the accuracy grows from 66.1% to 67.8% when the percentage of observed nodes goes up from 10% to 25% and becomes 70.0% for full observation of the cascade. Although such growth in accuracy is noticeable, it is not statistically significant, and there is a non-monotonic decline in accuracy on the 25% observation due to variance across folds (see Section 6.3). Such a finding clearly contradicts the findings of the initial pipeline, where equal values of accuracy, F1-score, and ROC-AUC for all levels of observation were considered “full performance preserved from minimal observation.” After careful examination, it was found out that such a result is based on a constant-output classifier that produces the same output whatever the input is. However, even a minimal observation of the cascade on 10% provides accuracy much higher than random and grows for more observations of the cascade.

6.5 Feature-Level Interpretation

As follows from the results of the correlation analysis discussed in Section 5.9, source credibility proves to be the most informative individual attribute ($r = 0.0956$). Also, several more affective attributes can be considered, as articles with admiration, excitement, approval, joy, pride, relief,

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and amusement attributes show a positive correlation with the Real class while neutral and disapproving attributes correlate with the Fake class. All the individual correlation coefficients are quite low in absolute values ($|r| < 0.10$). Such an observation corresponds well to the results of high discriminatory performance achieved in Sections 5.1–5.3, where the discriminative ability of the method relies on 100-dimensions PCA-based reduction of semantic embeddings and on a number of weak but informative emotions and credibility attributes. Finally, the expansion of the coverage of the source credibility attribute should be considered as the main way to improve classification performance since the attribute is highly informative, while it has quite a low coverage rate (allowing to align 4.8-7.9% of the articles with the reference domain).

6.6 Remaining Limitations

- Synthetic propagation cascades: As there is no genuine longitudinal social-media propagation cascade for the data set, the cascades are synthetically generated from article-level metadata based on a general empirical study regarding differential propagation of truthful versus deceptive information (Vosoughi et al., 2018). The cascades are not created based on any real observations. Therefore, the findings based on propagation and hybrid models presented in this thesis are indicative of the fact that the modeling technique suggested here is able to detect structure in time-ordered cascades in theory.
- Class balancing leading to a smaller corpus size: With the use of exactly 5,000 Fake and 5,000 Real documents (Section 3.2), a perfectly balanced classes system is attained despite the fact that most of the 17,441 Real-class documents are left unutilized. This means that the sacrifice made is the use of a smaller corpus in lieu of a balanced corpus, and other possible alternatives (such as class-weighting in the loss function across all the unbalanced corpus) have yet to be tried out.
- Even though the adjustment for scheme independence (Eq. (3.7)) improves the domain match score by five to eight times, most papers still display a neutral credibility value (greater than 90%), which seems too low to account for its importance.
- DetectGPT-lite continues to remain disabled because even though the AI text generation probability score algorithm works, it does not work by default since it used to give a constant value before (Section 3.6). The possible benefit that might arise from this tool if a scoring algorithm can be proved to work has not been examined.
- AI-image detector calibration: The independent image-detection task still demonstrates a high level of category dependency in terms of flag rates (0% in case of the Fruit category and 100% in case of Airplane and Car categories; see Section 3.8). This indicates that the threshold of detection needs to be calibrated separately for each category. Since the task is now independent of the article classification task, this problem does not influence the veracity-detection task.

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- Heterogeneity among file pairs: The second file pair is 8 to 10 points better than the first file pair and the whole database in terms of accuracy (refer to Section 5.3). This difference results from the relatively small size of the second file pair (862 papers).

The one split train/test set baseline comparison method is adopted. As discussed in Section 6.3, the assessment of the LSTM/CNN/static GNN/TGNN (Section 5.6) is done with one train/test split compared to the cross validation approach for the propagation model (Section 5.5).

6.7 Summary of Findings

Conclusion In conclusion, the present thesis has pinpointed and rectified two different structural flaws in an initial implementation of the Hybrid Fusion Framework: a mismatch between row-position-based feature extraction and labels, and an open time input leading to a breakdown of the propagation model. Having fixed those problems, above-chance performance across all sub-modules of the framework has been obtained after cross-validation. In particular, a content-only classifier has achieved 72.41% cross-validated accuracy, while a propagation-only classifier has gotten 65.75% cross-validated accuracy and has shown real, albeit very slight, early-detection capabilities, while the leak-free hybrid classifier has reached 72.58% accuracy and a significantly lower variance in comparison with both of the sub-modules. The results are fairly modest but not terrible, and the limitations of the framework which have been stated throughout the thesis include synthetical propagation data, lack of credibility data, turning off AI text detection, and a one-off split in comparison with the cross-validated architecture, which is scientifically correct.

6.8 Reconciling the Primary Protocol with the Full-Dataset Findings

Sections 6.2 and 5.12 appear, at first glance, to contradict each other: Section 6.2 concludes that content is the dominant signal and that stacking “mainly reduces variance”, based on 5-fold cross-validation and 70/30 hold-out evaluation over the full 9,995-article dataset, whereas Section 5.12's primary 70/15/15 protocol — restricted to the 1,200-article propagation-covered subset, with content, propagation, and hybrid scored on identical splits — finds propagation and hybrid decisively outperforming content. Both results are computed correctly; they answer different questions, and reporting both, rather than only the more flattering one, is the more honest choice for this thesis.

The key difference between the two protocols is sample size and comparison fairness. In Sections 5.1–5.2, the content branch is trained and evaluated on close to 7,000–9,995 articles, while the propagation branch (Sections 5.4–5.5) is necessarily restricted to the 1,200 articles for which synthetic cascades were generated (Section 4.1, MAX_ARTICLES_PER_CATEGORY_GRAPH = 600 per class). Comparing a content model trained on 9,995 articles against a propagation model trained on 1,200 cascades is not a matched comparison — the content branch simply has roughly eight times more training data. The primary 70/15/15 protocol and the 10-fold robustness check (Section 5.13) correct this by restricting content, propagation, and hybrid to the identical article

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subset and the identical splits within each repeat, so that any accuracy difference reflects the modality's information content rather than a training-set-size confound.

Once that confound is removed, the propagation signal turns out to be substantially more informative on this subset than the full-dataset comparison suggested, and the hybrid fusion is never worse than the stronger of its two constituent branches across any protocol reported in this thesis (Sections 5.7, 5.12, and 5.13). Two findings should therefore both be retained rather than one superseding the other: (i) on the full, largest-available dataset, a content-only classifier is a strong, data-efficient baseline that alone reaches 72–74% accuracy; and (ii) when compute allows the propagation branch to be trained on a comparably sized, matched subset, it closes — and on this dataset, reverses — the gap with content, and fusing the two never hurts and typically helps. The practical implication is that the value of the propagation branch in this framework is currently capped by how many synthetic cascades can affordably be generated (Section 4.1), not by any inherent weakness of the temporal-graph architecture itself; Section 7.2 lists relaxing this cap, and validating the finding on real (rather than synthetic) propagation cascades, among the priorities for future work.

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Chapter 7: Conclusion and Future Work

7.1 Conclusion

This thesis implements a design and evaluates a Hybrid Fusion Framework that can detect misinformation by modelling features like textual information, affective information, source credibility and social propagation regardless of whether the news article is written by a human or an AI system. The following six research questions (as stated in Section 1.3 (Scope of Study)) have all been fulfilled in relation to the design and evaluation of this framework: a feature extraction pipeline has been designed and used to extract features from a class-balanced dataset containing 9,995 news articles; a Time-Aware Graph Transformer has been developed and evaluated using 1,200 generated propagation cascades for a single-split diagnosis evaluation and a 5-fold cross-validation; a leakage-free out-of-fold stacking technique has been used to combine the predictions made by both content-level and propagation-level models; the resulting models have been evaluated against content-only baselines, sequence/graph-based baselines (LSTM, CNN, static GNN) and one particular early detection protocol.

Furthermore, the thesis illustrates how two separate but independently existing flaws leading to sub-chance accuracy and majority-class collapse of the previous pipeline design have been identified and resolved, namely: one is feature/label mismatch bug at the row level and the other one is unbounded temporal input for time2vec encoding in the propagation model. Therefore, all model designs considered exhibit consistent and significant performance above the 50% threshold for a nearly balanced binary classification problem. Best performing classifier based only on the content (a multilayer perceptron working with a 129-dimensional fused feature vector) shows 72.41% cross-validated accuracy and 74.12% hold-out (leak-free) accuracy; best performing propagation classifier demonstrates 65.75% cross-validated accuracy and gives a genuine early detection signal by boosting its performance from 66–68% for 10–25% of a cascade to 70.0% accuracy for a whole cascade; finally, the best hold-out (leak-free) ensemble of both classifiers shows 72.58% accuracy. Detailed numbers can be found in Chapter 5 while an honest interpretation is presented in Chapter 6. It is an important empirical contribution because it proves the capacity of the proposed system to find the meaning in the data after data integrity and numerical stability issues are taken into account, although the limitations of these findings exist because of synthetic nature of the propagation cascade and a small corpus size.

A later addition to the evaluation protocol (Section 5.12) refines this picture. When content, propagation, and hybrid are scored on identical 70/15/15 train/validation/test splits of the same 1,200-article propagation-covered subset — removing the training-set-size confound that favoured content in the full-dataset comparison above — the propagation branch reaches 77.22% accuracy and the hybrid fusion 77.89%, both clearly ahead of the best content-only candidate on that subset (67.78%); a 10-fold cross-validation robustness check (Section 5.13) confirms the same ordering and shows the hybrid's margin over content-only widening, not narrowing, as the fold count increases. Reconciling the two pictures (Section 6.8), this thesis concludes that content is a strong,

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data-efficient standalone baseline on the largest available sample, while propagation and hybrid fusion are the stronger choice once evaluated on a fairly matched, equal-sized comparison — and that the hybrid fusion, across every protocol tested in this thesis, never underperforms the better of its two constituent branches.

7.2 Recommendations for Future Work

- Realistic Propagation & Validation: Replace synthetic propagation graphs with real timestamped social-media reply graphs and apply the same 5-fold cross-validation across all models.
- Improved Features: Reintroduce the article-specific DetectGPT-lite feature and improve source-credibility classification using learned or continuously updated credibility signals.
- Imbalanced & Multimodal Data: Evaluate class-weighted training on the natural imbalanced dataset and improve image detection consistency while exploring real image–article multimodal datasets.
- Advanced Classification: Extend binary classification to multi-level veracity (true, partially false, false, satire) and include human-vs-AI authorship detection.
- Generalization & Architecture: Investigate cross-platform/cross-lingual performance and replace the two-stage stacking method with an end-to-end jointly trained content–propagation network.

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